

Question 1

Consider the fully recurrent network architecture (without output activation and bias units) defined as

$$s(t) = Wx(t) + Ra(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$ and parameter matrices R, W, V . Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\delta(t) = \frac{\partial L}{\partial s(t)}$ is a column vector. Which of the following statements are true?

- ☒ a. The deltas fulfill the recursive relation $\delta(t) = \text{diag}(f'(s(t))) (V^T \frac{\partial L}{\partial \hat{y}(t)} + R^T \delta(t+1))$
- b. The gradients for the recurrent weights R can be written as $\sum_{t=1}^T \delta(t) s^T(t-1)$
- ☒ c. The gradients for the input weights W can be written as $\sum_{t=1}^T \delta(t) x^T(t)$
- d. BPTT is an algorithm to compute gradients that are local in time.
- e. The memory demand for BPTT is quadratic in the sequence length T .

Question 2

Which of the following statements about RNNs are true?

- a. The number of RNN parameters grows linearly with the sequence length.
- b. Many RNN architectures are Turing complete, which means that they can learn to exactly imitate the behavior of every computer program.
- ☒ c. RNNs can be applied to input sequences of varying lengths.
- ☒ d. An RNN is equivalent to a feedforward network applied to the input and an internal state representation that depends on previous time steps.
- e. An RNN is equivalent to a feedforward network applied to the input at every time step separately.

Question 3

The Elman Network (without output activation function and bias units) can be defined as

$$s(t) = Wx(t) + a(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$, and parameter matrices W, V . Which of the following statements are true?

- ☒ a. The recurrent weights are constant.
- ☒ b. The recurrent weights are equal to the identity matrix.
- c. The hidden units of the Elman network are interconnected in the sense that they can exchange information at every time step.
- ☒ d. Elman proposed to train this network locally in time, i.e., on each time step separately.
- e. The recurrent Jacobian of the hidden activations is $\frac{\partial a(t)}{\partial a(t-1)} = \text{diag}(f'(s(t-1)))$.

Question 4

The LSTM (without bias units and forget gate) is defined as

$$i(t) = \tanh(Ux(t) + Ph(t - 1))$$

$$i(t) = \sigma(Vx(t) + Qh(t - 1))$$

$$c(t) = c(t - 1) + z(t) \odot i(t)$$

$$o(t) = \sigma(Wx(t) + Rh(t - 1))$$

$$h(t) = \tanh(c(t)) \odot o(t)$$

with input vectors $x(t)$, hidden activation vectors $h(t)$, memory cell state vectors $c(t)$, gate activation vectors $z(t)$, $i(t)$, $o(t)$, weight matrices P, Q, R, U, V, W . Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial c(t)}$ is a column vector. Which of the following statements are true?

- a. The gradient of the loss with respect to the hiddens is $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t)}{\partial h(t)} + U \frac{\partial L}{\partial z(t)} + V \frac{\partial L}{\partial i(t)} + R \frac{\partial L}{\partial o(t)}$, but it was originally proposed to be approximated by $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t)}{\partial h(t)}$
- ☒ b. The memory cell fulfills $\frac{\partial L}{\partial c(t-\tau)} = I$ for any $\tau \in \{1, \dots, t-1\}$, where I is the identity matrix. This solves the vanishing gradient problem and is called constant error carousel.
- c. Because of the simple structure of the memory cell, the LSTM architecture fails to be Turing complete.
- d. The gradient of the loss with respect to the memory cell is $\frac{\partial L}{\partial c(t)} = \frac{\partial L}{\partial c(t-1)} + o(t) \odot \tanh'(c(t)) \odot \frac{\partial L}{\partial h(t)}$
- e. The LSTM architecture has no exploding or vanishing gradients because $\frac{\partial h(t)}{\partial h(t-1)}$ is an orthogonal matrix.
- ☒ f. If we choose $z(t) = \sigma(Ux(t) + Ph(t - 1))$, then $E[z(t)] > 0$ and the memory cells will always increase at every time step. This can be a problem for very long sequences but can also be helpful for certain problems.

Question 5

Which statements about LSTM are true?

- ☒ a. Most LSTM networks used in practice feature a forget gate.
- ☒ b. The focused LSTM network feeds the input vector $x(t)$ only into the block input $z(t)$ and not into input gate $i(t)$ or output gate $o(t)$.
- c. To counter the drift effect, the input gate should be initialized with positive values.
- ☒ d. "Ticker steps" refers to the technique of letting an LSTM run for more time steps even after the input sequence has been taken in by the network.
- e. The forget gate should be initialized with negative bias values.
- f. The inputs to the lightweight LSTM have to fulfill the Markov property.
- ☒ g. An LSTM with coupled input gate and forget gate, i.e., $c(t) = (1 - i(t)) \odot c(t - 1) + i(t) \odot z(t)$, leads (alongside other changes) to the gated recurrent unit (GRU).
- h. The forget gate solves the vanishing gradient problem.

Question 6

Consider the fully recurrent network architecture (without output activation and bias units) defined as

$$s(t) = Wx(t) + Ra(t - 1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$ and parameter matrices R, W, V . Let θ denote the vector of all network parameters shared in time and let $\theta(t)$ denote their usage at time t . Further, let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial \theta}$ is a column vector.

Which statements about RTRL are true?

- a. Schmidhuber's approach divides the input sequence into chunks of size N (number hidden units) and performs RTRL within these chunks. Then it uses BPTT to consolidate the gradients for these chunks.

b. BPTT considers the recursion $\frac{\partial L}{\partial s(t)} = \frac{\partial s(t+1)}{\partial s(t)} \frac{\partial L}{\partial s(t+1)} + \frac{\partial L}{\partial s(t)}$ while RTRL considers the recursion $\frac{\partial s(t)}{\partial \theta} = \frac{\partial s(t)}{\partial \theta(t)} + \frac{\partial s(t)}{\partial s(t-1)} \frac{\partial s(t-1)}{\partial \theta}$.

c. RTRL computes the gradients $\frac{\partial L(t)}{\partial \theta}$ during the forward pass already.

- d. The term $\frac{\partial s(t)}{\partial \theta}$ generally has $O(N^4)$ elements, where N denotes the number of hidden units

Question 7

Backpropagation through time (BPTT) is the standard gradient computation algorithm for RNNs. However, there exist alternatives, e.g., truncated BPTT and real-time recurrent learning (RTRL). Which statements about these algorithms are true?

- a. Truncated BPTT has advantages if the input sequences are short.

- b. Truncated BPTT computes the same gradients as BPTT.

c. RTRL is very slow and therefore rarely used in practice.

- d. The runtime complexity of RTRL is $O(N^3)$ per time step, where N denotes the number of hidden units.

e. RTRL computes the same gradients as BPTT

f. Truncated BPTT divides the input sequence into chunks and propagates the deltas only within chunks but not across them.